Sebastian G. Gruber



Machine Learning Engineer & Data Scientist

PhD in Machine Learning | MSc in Statistics

Frankfurt am Main, Germany

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PROFILE

Machine learning engineer and statistician by training (BSc/MSc Statistics, PhD in machine learning, summa cum laude) who builds data-driven systems end to end – from data generation, validation and statistical analysis to fine-tuned models, cloud pipelines and product integration. Currently at LINK2AI, taking an LLM hallucination detector from first experiments to a GPU-served module of the company's LLM monitoring platform. Particular strengths: uncertainty quantification and calibration, rigorous evaluation, and understanding how the data-generating process shapes what a model learns. First-author papers at NeurIPS, ICML, AISTATS and UAI; 300+ citations.

SKILLS

Statistics: statistical modelling & inference, regression, bootstrap confidence intervals, hypothesis testing, confounder analysis, uncertainty quantification & calibration, proper scoring rules, inter-annotator agreement

Machine learning: supervised learning (e.g. random forests), deep learning, fine-tuning of LLMs and encoders (LoRA/QLoRA, 4-bit quantisation), LLM evaluation & LLM-as-a-judge, RAG, embeddings, synthetic data generation

Programming: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, Hugging Face Transformers/PEFT, vLLM), R (mlr3), SQL, Julia

MLOps & engineering: AWS, Terraform, Docker, Ray Serve, MLflow, Optuna, GitLab CI, FastAPI, Streamlit, Linux, Git, pytest; OpenAI/Anthropic APIs, LiteLLM; AI-assisted development (Claude Code)

EXPERIENCE

Machine Learning Engineer — Feb 2026 – present

LINK2AI GmbH — Frankfurt am Main

Start-up for the security and trustworthiness of LLM applications (runtime monitoring and protection platform).

- **Hallucination detection for RAG:** owned the answer-grounding sensor end to end within an ESA BIC Hessen incubation project – problem framing, data, evaluation, model selection and product integration – to flag LLM answers that are not supported by the retrieved sources.
- **Evaluation & statistics:** built a reproducible benchmarking framework on public and domain-specific data (per-class error rates, AUROC, latency percentiles, MLflow tracking); used partial correlations, fixed-effects regression and bootstrap confidence intervals to identify what drives performance, showing that how a dataset was constructed – its data-generating process – largely determines what models learn from it.
- **Data:** created synthetic, domain-specific evaluation sets from technical documents, labelled by a multi-vendor LLM-judge panel and validated against blind human annotation (Cohen's κ); uncovered substantial label noise in public benchmarks.
- **Modelling:** fine-tuned and compared ModernBERT encoders and 4-bit LoRA-adapted LLMs (Qwen, Mistral, Gemma) against NLI, embedding, open-source detector and commercial LLM-judge baselines; an on-premise fine-tuned model is competitive with commercial LLM judges at an order of magnitude lower latency, and per-deployment threshold tuning proved the largest single lever.
- **Product integration:** implemented the detector as a GPU-served module of the company's monitoring platform (PyTorch, Ray Serve, 4-bit inference) with structure-aware chunking of long documents, hardened parsing of uploaded PDF/Office/HTML files, a configuration UI (Next.js/React), GPU tests in CI and documentation.
- **Policy classifiers & synthetic data:** built a pipeline that turns a chatbot's compliance policy into one fine-tuned lightweight classifier per rule – LLM-generated training data with diversity control (own Python library: Vendi score and spectral topic extraction on embeddings), Optuna sweeps tracked in MLflow, export to the Hugging Face Hub and serving via vLLM.

Postdoctoral Researcher in Machine Learning — Apr 2025 – Dec 2025

KU Leuven, ESAT-PSI (Prof. Matthew Blaschko) — Leuven, Belgium / remote

- Uncertainty calibration for trustworthy AI, e.g. confidence-aware estimation of medical-imaging biomarkers (MIDL 2026); co-supervised PhD students, resulting in three co-authored papers.
- Co-organised the AISTATS 2026 workshop *Towards Trustworthy Predictions: Calibration for Modern AI* with researchers from Inria, Microsoft Research and UPenn (33 accepted papers); NeurIPS 2025 Top Reviewer.

Doctoral & Postdoctoral Researcher in Machine Learning Mar 2021 – Mar 2025
German Cancer Research Center (DKFZ), ML in Oncology group (Prof. F. Buettner) Frankfurt am Main

- Developed a statistical framework for quantifying the uncertainty of ML predictions (proper scores, bias–variance decompositions, calibration-error estimators), from classifiers to generative models and LLMs; PhD summa cum laude based on six peer-reviewed papers.
- Three-month research visit at Inria Paris (Francis Bach), funded by a DKFZ travel grant; resulted in a TMLR paper.
- Supervised eight theses (seven BSc, one MSc); lectures, tutorials and exams for *Introduction to Methods of AI* at Goethe University (2021–2024).
- Nine conference and invited talks, incl. Bayer AG Research (2022) and the Helmholtz AI Conference (2024).

Working Student & Master's Thesis, Machine Learning Aug 2018 – Feb 2021
Siemens AG Munich

- **Production ML pipeline** (IT Analytics Lab): built a live prediction service on AWS, provisioned with Terraform, that queries a company database every 10 minutes, scores new records with a random-forest model and writes the predictions back – starting automatically every workday morning and shutting down in the evening to save cloud costs.
- **Research** (Corporate Technology, Machine Intelligence): post-hoc uncertainty calibration under domain shift, resulting in a CVPR 2021 paper (130+ citations) and a US patent application; master's thesis on meta-learning and knowledge distillation.

Working Student, Data Analytics Jul 2017 – Jul 2018
Criteo Munich

- Designed and built a web application that automatically generates quarterly business reports.

EDUCATION

PhD in Computer Science (Dr. rer. nat.) – summa cum laude Mar 2021 – Jan 2025
Goethe University Frankfurt Frankfurt am Main
Thesis: *A Novel Framework for Uncertainty Quantification via Proper Scores for Classification and Beyond*

MSc in Statistics – grade 1.4 Oct 2018 – Feb 2021
Ludwig Maximilian University (LMU) Munich
Thesis on meta-learning and knowledge distillation (with Siemens Corporate Technology)

BSc in Statistics – grade 1.43 Oct 2015 – Sep 2018
Ludwig Maximilian University (LMU) Munich
Thesis on reinforcement learning

SELECTED PUBLICATIONS

- **Gruber, S. G.**, Walha, N., Bach, F., Buettner, F. (2026). *Eigenvalue Calibration for Semantic Embeddings of Large Language Models.* UAI.
- Walha, N., **Gruber, S. G.**, et al. (2026). *Fine-grained Uncertainty Decomposition in Large Language Models: A Spectral Approach.* AAAI.
- **Gruber, S. G.**, Buettner, F. (2024). *A Bias-Variance-Covariance Decomposition of Kernel Scores for Generative Models.* ICML.
- Popordanoska, T.*, **Gruber, S. G.***, et al. (2024). *Consistent and Asymptotically Unbiased Estimation of Proper Calibration Errors.* AISTATS.
- **Gruber, S. G.**, Buettner, F. (2022). *Better Uncertainty Calibration via Proper Scores for Classification and Beyond.* NeurIPS.
- Tomani, C., **Gruber, S. G.**, et al. (2021). *Post-hoc Uncertainty Calibration for Domain Drift Scenarios.* CVPR (oral).

*equal contribution. 11 papers and one US patent application (17/347,645); 300+ citations – Google Scholar.

ADDITIONAL

Open source: mlr3hyperband (CRAN package for hyperparameter optimisation in mlr3); mlrPlayground (interactive web app teaching ML algorithms)
Awards: PhD summa cum laude; NeurIPS 2025 Top Reviewer; DKFZ short-term travel grant (2024)
Languages: German (native), English (fluent), French (basic)